

Chapter 5

Transform Analysis of Linear Time-Invariant Systems

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5.1 System Functions with LCCDE Representation

The class of systems with input/output relation

$$\sum_{k=0}^N a_k y[n-k] = \sum_{k=0}^M b_k x[n-k]$$

and zero initial condition is linear and time-invariant.

5.1 LCCDE System Function

- If we take the z -transform of both sides using linearity and time-invariance, we can write

$$\begin{aligned} H(z) &= \frac{\sum_{k=0}^M b_k z^{-k}}{\sum_{k=0}^N a_k z^{-k}} \\ &= \frac{b_0 \prod_{k=1}^M (1 - c_k z^{-1})}{a_0 \prod_{k=1}^N (1 - d_k z^{-1})}. \end{aligned}$$

- Each numerator factor $(1 - c_k z^{-1})$ contributes a zero at $z = c_k$ and a pole at $z = 0$.
- Each denominator factor $(1 - d_k z^{-1})$ contributes a pole at $z = d_k$ and a zero at $z = 0$.
- If $H(z)$ is causal then $h[n]$ must be a right-sided sequence and thus the ROC of $H(z)$ must be the exterior of a circle with maximum pole radius.

Example 5.1: A Second-order LCCDE

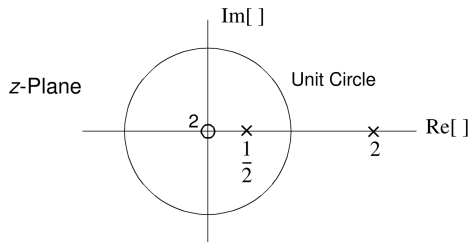
An LTI system has input/output relationship

$$y[n] - \frac{5}{2}y[n-1] + y[n-2] = x[n].$$

- Taking the z -transform of both sides we get

$$H(z) = \frac{1}{(1 - \frac{1}{2}z^{-1})(1 - 2z^{-1})}.$$

Example 5.1: Pole-zero Plot and ROC



Pole-zero plot of $H(z)$.

There are three possible choices for the ROC:

- If the system is to be causal, $|z| > 2$, but then $H(z)$ is not stable.
- The system is stable if we choose $\frac{1}{2} < |z| < 2$, but then the system cannot be causal.
- If we choose $|z| < \frac{1}{2}$, the system is neither stable nor causal.

5.2 Inverse Systems

For any LTI system with system function $H(z)$, the corresponding system inverse, $H_i(z)$, is defined such that

$$H(z)H_i(z) = 1$$

thus

$$H_i(z) = \frac{1}{H(z)}.$$

- The frequency response, $H_i(e^{j\omega})$, may not exist, e.g. consider an ideal lowpass filter; frequency components above ω_c are set to zero by $H(e^{j\omega})$ and thus cannot be recovered.
- Systems with rational system functions have inverses of the form

$$H_i(z) = \left(\frac{a_0}{b_0}\right) \frac{\prod_{k=1}^N (1 - d_k z^{-1})}{\prod_{k=1}^M (1 - c_k z^{-1})}.$$

5.2 ROC and Stable Inverses

- Poles of $H(z)$ imply zeros of $H_i(z)$.
- Zeros of $H(z)$ imply poles of $H_i(z)$.
- The ROC of $H_i(z)$ must overlap the ROC of $H(z)$, thus if $H(z)$ is causal the ROC of $H_i(z)$ must overlap

$$|z| > \max_k |d_k|.$$

- If $H(z)$ is also stable and we further require $H_i(z)$ to be stable, then both the poles and zeros of $H(z)$ must lie within the unit circle.

Example 5.2: A First-order System

Let

$$H(z) = \frac{1 - 0.5z^{-1}}{1 - 0.9z^{-1}}, \quad \text{ROC} = |z| > 0.9.$$

- By definition

$$H_i(z) = \frac{1 - 0.9z^{-1}}{1 - 0.5z^{-1}}$$

with $\text{ROC} = |z| > 0.5$, which overlaps with $|z| > 0.9$.

- Inverse transforming we get

$$h_i[n] = (0.5)^n u[n] - 0.9(0.5)^{n-1} u[n-1],$$

which is both stable and causal.

5.3 Impulse Response for LTI Systems

First-Order Poles

A partial fraction expansion for rational $H(z)$, with only first-order poles, can be written as

$$H(z) = \sum_{r=0}^{M-N} B_r z^{-r} + \sum_{k=1}^N \frac{A_k}{1 - d_k z^{-1}}.$$

- Inverse transforming term-by-term, assuming causality, yields

$$h[n] = \sum_{r=0}^{M-N} B_r \delta[n - r] + \sum_{k=1}^N A_k d_k^n u[n].$$

- If $h[n]$ contains at least one term of the form $A_k(d_k)^n u[n]$, then the system is called an infinite-impulse response (IIR) system.

5.3 No Poles Except at $z = 0$

- If $H(z)$ contains no poles, except at $z = 0$, then $N = 0$ and

$$H(z) = \sum_{k=0}^M b_k z^{-k}.$$

- The impulse response is simply

$$h[n] = \sum_{k=0}^M b_k \delta[n - k],$$

and the system is called a finite-impulse response (FIR) system.

- For FIR systems the difference equation can be represented as

$$y[n] = \sum_{k=0}^M b_k x[n - k],$$

which is identical to the convolution sum formula.

5.4 Frequency Response for LTI Systems

We are interested in the magnitude, phase, and group delay associated with $H(e^{j\omega})$.

- In terms of the poles and zeros of $H(z)$ we can write

$$H(e^{j\omega}) = \left(\frac{b_0}{a_0} \right) \frac{\prod_{k=1}^M (1 - c_k e^{-j\omega})}{\prod_{k=1}^N (1 - d_k e^{-j\omega})}.$$

- The magnitude response may be expressed as

$$|H(e^{j\omega})| = \left(\frac{b_0}{a_0} \right) \frac{\prod_{k=1}^M |1 - c_k e^{-j\omega}|}{\prod_{k=1}^N |1 - d_k e^{-j\omega}|}.$$

5.4 Gain, Attenuation, and Phase

- Frequently we may wish to consider the gain in dB or attenuation in dB of $H(e^{j\omega})$:

$$\text{Gain in dB} = |H(e^{j\omega})|_{\text{dB}} = 20 \log_{10} |H(e^{j\omega})|,$$

$$\text{Attenuation in dB} = -\text{Gain in dB}.$$

- The phase response can be written as

$$\angle H(e^{j\omega}) = \angle \left[\frac{b_0}{a_0} \right] + \sum_{k=1}^M \angle [1 - c_k e^{-j\omega}] - \sum_{k=1}^N \angle [1 - d_k e^{-j\omega}].$$

- The principal value of the phase of $H(e^{j\omega})$ is denoted by $\text{ARG}[H(e^{j\omega})]$, where

$$-\pi < \text{ARG}[H(e^{j\omega})] \leq \pi.$$

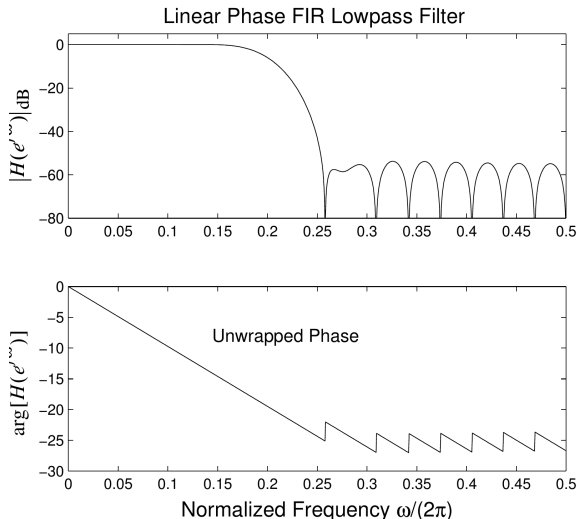
5.4 Wrapped and Unwrapped Phase

- In general $\text{ARG}[H(e^{j\omega})]$ contains $\pm 2\pi$ phase jumps, whereas $\arg[H(e^{j\omega})]$ is by definition continuous on $(0, \pi)$.
- To convert $\text{ARG}[H(e^{j\omega})]$ to $\arg[H(e^{j\omega})]$ we may write

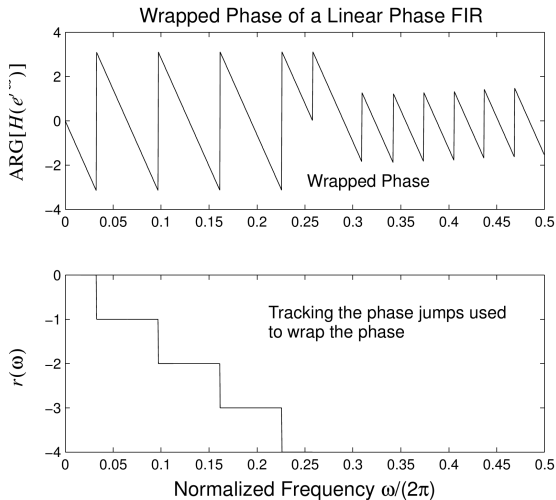
$$\arg[H(e^{j\omega})] = \text{ARG}[H(e^{j\omega})] + 2\pi r(\omega),$$

where $r(\omega)$ is an integer-valued function.

The figure shows the magnitude and unwrapped ($\arg[H(e^{j\omega})]$) phase of a linear phase LPF.

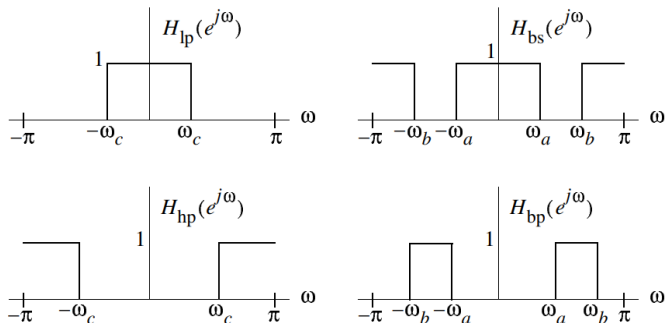


5.4 Wrapped Phase and Integer Function $r(\omega)$



The corresponding wrapped phase ($\text{ARG}[H(e^{j\omega})]$) along with integer function $r(\omega)$ for the FIR LPF.

5.5 Ideal LTI Systems: Frequency Selective Filters



Ideal filters over the fundamental period: (a) lowpass, (b) highpass, (c) bandstop, and (d) bandpass.

- The phase response is taken to be zero.
- Ideal filters have impulse responses extending from $-\infty$ to $+\infty$, thus making them computationally unrealizable.

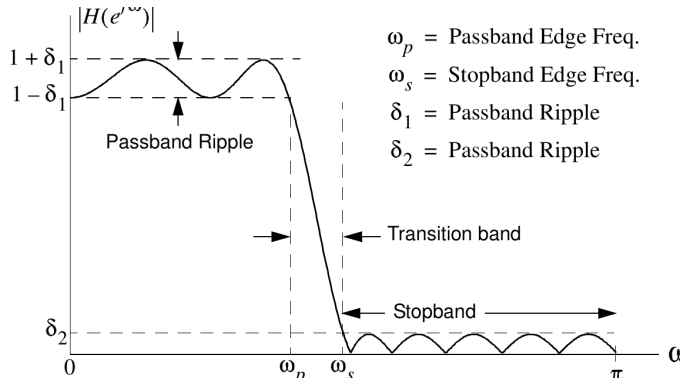
5.5 Paley-Wiener Theorem

If $h[n]$ has finite energy and $h[n] = 0$ for $n < 0$, then

$$\int_{-\pi}^{\pi} \left| \ln |H(e^{j\omega})| \right| d\omega < \infty.$$

- An important conclusion from the Paley-Wiener theorem is that $|H(e^{j\omega})|$ may be zero at a finite number of frequencies, but it cannot be zero over a band of frequencies without the integral becoming infinite. Any ideal filter will therefore be noncausal.
- It is nevertheless possible to design realizable filters that approximate ideal filters.

5.5 Realizable Frequency Selective Filters



Magnitude response of a realizable lowpass filter.

- A realizable lowpass filter includes a passband, a transition band, and a stopband.
- The passband and stopband are specified with ripple tolerances rather than ideal brick-wall transitions.

5.5 Phase Distortion and Pure Delay

Consider an ideal delay system with impulse response

$$h_{id}[n] = \delta[n - n_d].$$

- The frequency response is

$$H_{id}(e^{j\omega}) = e^{-j\omega n_d} = 1 \angle (-\omega n_d), \quad |\omega| < \pi,$$

assuming 2π periodicity.

- Linear phase which results from a pure delay is considered an acceptable form of distortion in most cases.

5.5 Ideal Lowpass with Linear Phase

For example, an ideal lowpass filter with linear phase can be defined as

$$H_{lp}(e^{j\omega}) = \begin{cases} e^{-j\omega n_d}, & |\omega| < \omega_c, \\ 0, & \omega_c < \omega \leq \pi, \end{cases}$$

and the impulse response written as

$$h_{lp}[n] = \frac{\sin(\omega_c(n - n_d))}{\pi(n - n_d)}, \quad -\infty < n < \infty.$$

- Note that an ideal lowpass filter is still unrealizable (noncausal) no matter how large we make n_d .
- A measure of the linearity of the phase response is the *group delay*.

5.5 Definition of Group Delay

- Formally, group delay is defined as

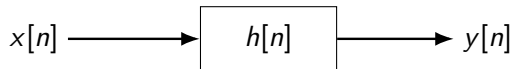
$$\tau(\omega) = \text{grd}[H(e^{j\omega})] = -\frac{d}{d\omega} \{ \arg[H(e^{j\omega})] \}$$

- The deviation of $\tau(\omega)$ from a constant indicates the degree of the nonlinearity of the phase.

Summary: Time-domain Characterization

Up to this point we have used the convolution, Fourier transform and z-transform as tools for the analysis of discrete-time systems.

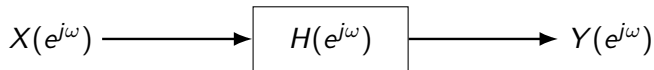
- In the time domain, an LTI system is completely characterized by its impulse response $h[n]$.



$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k].$$

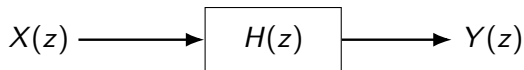
Summary: Transform-domain Characterization

- The Fourier transform of $h[n]$, if it exists, gives the *frequency response* as a complete characterization of an LTI system.



$$Y(e^{j\omega}) = H(e^{j\omega})X(e^{j\omega}).$$

- The z-transform of $h[n]$, within the appropriate region of convergence, gives the *system function* as a complete characterization of an LTI system.



$$Y(z) = H(z)X(z).$$